SIXT HACKATHON 2026 · PROJECT 2 · THE GOLDEN BUTTERFLY

One engine that forecasts every car's mileage and catches wrong reports

For cars whose last reading is weeks old (median 67 days to the target date)

TASK A

Forecast

Predicts km/day from what SIXT knows today, any date up to ~5 months ahead, with an 80% range

1,416 km

MAE, −16% vs best baseline

TASK B

Detect

Checks every report against the readings around it and the forecast, suggests the fix

5x

fewer alarms, same catch rate

Tested on vehicles never seen in training · no labels needed · runs on a laptop

APPROACH

Predict km per day from what is known today, trained by replaying the past

km(t) = last reading + predicted km/day × days since that reading

Forecast · LightGBM

- Learns km/day from what is known at the cutoff
- Trained by replaying the past: a pretend “today” every 30 days
- Tested on vehicles never seen in training

Detection · same expectations

- Physical rules
- Check against the readings before and after
- Real-time mode using the forecast

KEY INSIGHT: BUSY CARS REPORT MORE OFTEN

29 vs 83

median km/day, cars with ≤ 10 vs 31+ report days (Jul–Nov 2025)

Weighting each car by its true number of targets fixed a **−400 km bias** (−430 → −194 km)

No leakage: actual return dates not used · test data never used for training

FORECAST RESULTS · HELD-OUT VEHICLES

16% lower error than the best naive baseline

1,416 km

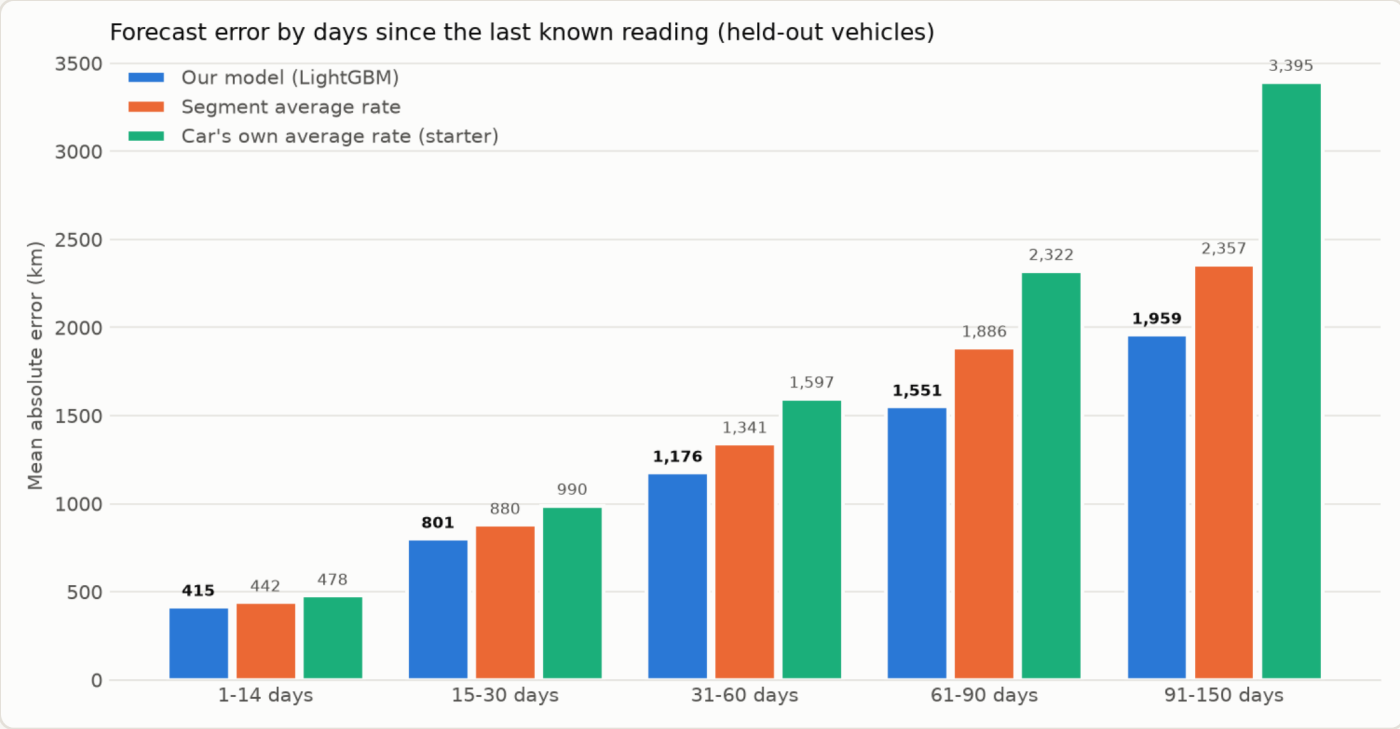
mean absolute error (MAE)

−16% vs segment average (1,680)
−36% vs starter baseline (2,200)

−48% when the last reading is older than 28 days (monthly reporting)

82% real coverage of our 80% uncertainty bands

3.6% error on the total km of the whole held-out fleet



The gain holds at all 3 validation periods (−12% to −16%) and grows with the horizon.

ERRONEOUS MILEAGE DETECTION · VS THE STARTER RULES, ON PLANTED ERRORS

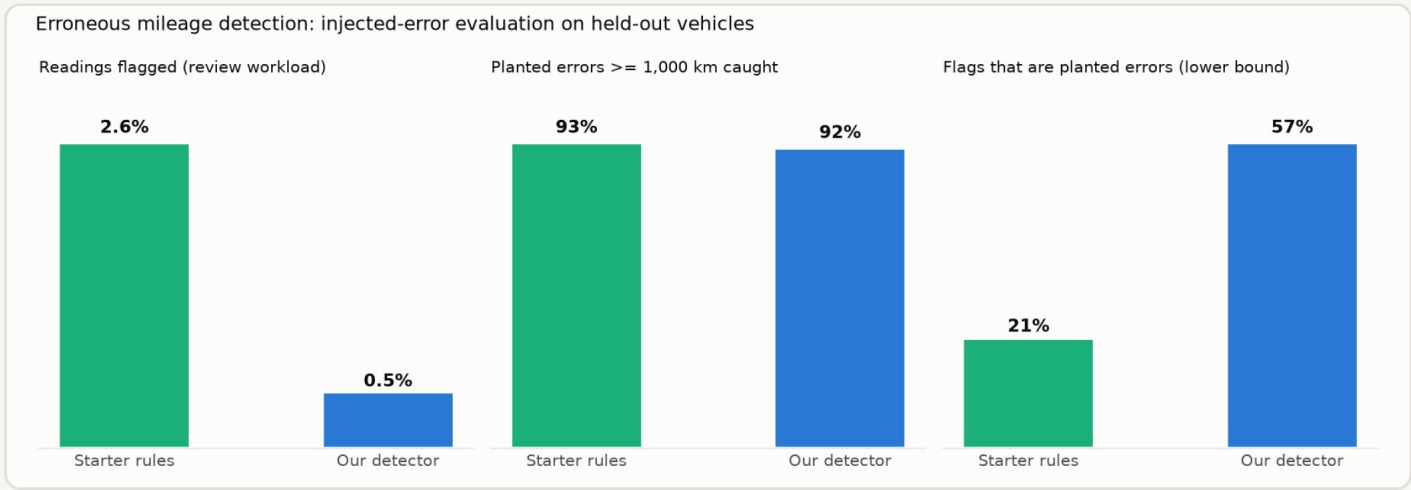
Same catch rate, 5x fewer alarms

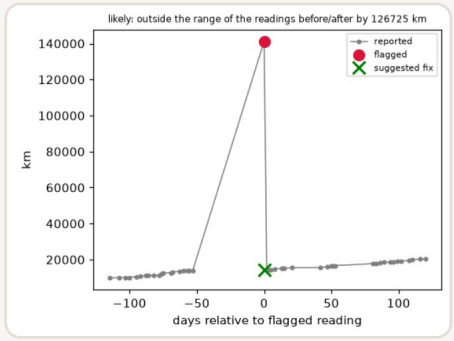
0.5%

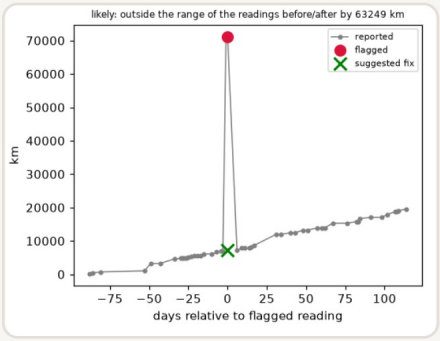
of readings flagged, vs 2.6% for the starter rules

92% of errors ≥ 1,000 km caught (starter rules: 93%)

97–98% right value suggested for swapped digits and miles-as-km







ORGANISERS' TEST DATA

1,018 flags

551,098 km between 55,074 and 55,412 → fix **55,109**

LIMITS AND NEXT STEPS

The missing signal is future rental activity, and SIXT already has it

Honest limits

- The remaining error is future rental activity: a linear model is only 5% behind, ensembles add 0.3%
- Stuck odometers can't be detected from mileage alone
- Test predictions are unscored

Next steps

- Plug in SIXT's booking calendar
- Review queue whose decisions become labels
- Request check-in readings from the most uncertain cars
- Score every incoming report in real time

ASK WHERE IT MATTERS

8.4%

of total error removed by a 60-day check-in from the 20% widest-band cars, vs 5.8% for 20% random cars